

Proof of Concept

Agentic Ethics Reasoning System

A Voice-Controlled, Multi-Model AI for Ethical Deliberation

Interim Assignment Submission
Ethics in Artificial Intelligence
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Part 1: Description of the System

What Is It?

The proposed final project is a locally deployed, voice-controlled agentic artificial intelligence system designed to reason through ethical dilemmas and produce actionable recommendations. Rather than relying on a single monolithic model, the system employs a multi-agent architecture in which several specialized large language models—each fine-tuned for a distinct reasoning task—collaborate under the direction of a central orchestrator. The user interacts with the system entirely through natural speech: posing questions aloud, receiving audible responses, and engaging in a back-and-forth Socratic dialogue about the ethical dimensions of a given concern.

What Need Does It Address?

The modern digital landscape is rife with ethically ambiguous technologies—dark-pattern interfaces, exploitative data-harvesting schemes, algorithmic bias in hiring or lending tools, and platforms that amplify misinformation. Ordinary users lack the philosophical vocabulary or structured reasoning frameworks to articulate why something feels wrong, let alone to take meaningful corrective action. This system addresses that gap by serving as an ethical co-reasoner: a tool that meets people where they are (conversational speech) and walks them through rigorous ethical analysis drawn from the accumulated traditions of moral philosophy.

How Does It Work? The Three-Phase Pipeline

The system follows a three-phase pipeline for every ethical query, each phase corresponding to a specialized agent:

- I. **Ethical Discourse.** The Dialogue Agent engages the user in a Socratic-style conversation, drawing on canonical ethical frameworks—utilitarian calculus, Kantian deontology, virtue ethics, care ethics, and the course’s central concept of decency—to help the user surface and articulate the moral stakes of their concern. For example, if a user asks whether a particular website’s use of dark patterns is ethical, this agent would walk the user through questions about informed consent, user autonomy, and the proportionality of benefit versus harm.
- II. **Ethical Analysis.** The Ethics Reasoner Agent takes the structured output of the conversation and applies formal ethical reasoning. It weighs the issue against multiple philosophical lenses, identifies conflicts between frameworks (e.g., a utilitarian outcome that violates a deontological duty), and produces a cogent, reason-based ethical verdict. The user receives not just a yes-or-no judgment but a transparent chain of reasoning they can evaluate for themselves.
- III. **Ethical Action.** The Action Agent translates the ethical verdict into practicable countermeasures. Depending on the scenario, this could mean drafting a complaint letter, generating a browser-extension script to block dark-pattern elements, composing a social-media awareness post, or producing a structured

report suitable for submission to a regulatory body. The system does not merely opine—it helps the user do something about it.

Example Scenario

User Query	<i>"Is this e-commerce website ethical? It makes it almost impossible to unsubscribe from their mailing list."</i>
Phase I Output	Socratic dialogue exploring consent, autonomy, the concept of roach motels in UX, and the asymmetry of entry vs. exit in platform design.
Phase II Output	Ethical verdict: the practice violates Kantian respect for autonomy, fails a utilitarian analysis (net harm outweighs company benefit), and offends the standard of decency by treating users as means rather than ends.
Phase III Output	Generates a direct-unsubscribe URL, drafts an FTC complaint template, and produces a concise social-media post warning other consumers.

Part 2: Plan of Construction

Technology Stack

The system will be built using the following tools and technologies:

Component	Tool / Framework
Development IDE	Claude Code (Anthropic’s agentic coding tool) for building, debugging, and iterating on the codebase
Language Models	HuggingFace Hub — multiple open-source LLMs selected and fine-tuned for specific reasoning modalities (e.g., Mistral, Phi, LLaMA family)
Agent Orchestration	Python with LangChain or a custom lightweight agent framework to coordinate multi-model pipelines
Speech-to-Text	OpenAI Whisper (open-source) running locally for privacy-preserving voice transcription
Text-to-Speech	Coqui TTS or Bark (open-source) for natural-sounding voice output
Local Deployment	Ollama or vLLM for efficient local model serving; Docker containerization for reproducibility
Fine-Tuning	QLoRA / PEFT techniques via the HuggingFace Transformers library for parameter-efficient fine-tuning on ethics-specific datasets
Frontend (optional)	Gradio or Streamlit for a minimal web-based interface alongside the voice interface

Development Roadmap

Development will proceed in four iterative phases:

- Foundation (Weeks 1–2):** Set up the local model-serving infrastructure using Ollama. Select candidate base models from HuggingFace (prioritizing models with strong instruction-following capabilities in the 7B–13B parameter range for local feasibility). Implement the basic voice pipeline: Whisper for input, a placeholder TTS for output.
- Agent Architecture (Weeks 2–3):** Build the orchestrator agent that routes user queries through the three-phase pipeline. Implement inter-agent communication protocols. Design the structured prompting templates that encode ethical frameworks into each agent’s system instructions.
- Fine-Tuning (Weeks 3–4):** Curate training data from ethical philosophy corpora, case-study databases, and annotated moral dilemmas (e.g., the Moral

Machine dataset, the ETHICS benchmark). Apply QLoRA fine-tuning to specialize each model for its role.

4. **Integration & Polish (Week 5):** End-to-end testing with real ethical scenarios. Refine the TTS output for natural conversational cadence. Build the optional Gradio/Streamlit UI. Prepare the video demonstration.

Sharing and Demonstration

The system will be demonstrated via a 10–15 minute video presentation showing live interaction with the voice interface. All source code will be made available through a GitHub repository or a compressed archive submitted alongside the video. Because the system runs entirely on local hardware, the instructor and fellow students can clone the repository and run the system themselves provided they have access to a machine with a compatible GPU (or sufficient CPU resources for smaller quantized models). A Docker Compose configuration will be provided to simplify setup.

Part 3: Ethical Design Framework

Decency as a Design Principle

The central ethical commitment of this project is the concept of decency as explored throughout this course. Decency, in this context, is not merely the absence of harm but the active orientation of a technology toward the dignity, autonomy, and well-being of every person it touches. Every design decision in this system has been filtered through that standard. The system does not make ethical decisions for the user; it reasons with them. It does not impose a single philosophical dogma; it presents multiple frameworks and helps the user weigh them. It does not merely critique; it empowers the user to act. This tripartite structure—dialogue, analysis, action—is itself an expression of decency: treating the user as a rational moral agent capable of arriving at their own conclusions when given the proper tools.

Steps Taken to Ensure Ethical Use

Several deliberate design choices embed ethical safeguards into the system's architecture:

- **Local-First Privacy:** All models run on the user's own hardware. No voice data, transcriptions, or ethical queries are transmitted to external servers. This eliminates the surveillance risk inherent in cloud-based AI assistants and respects the user's right to private moral deliberation—a space that should be as inviolable as the confessional or the therapist's office.
- **Philosophical Pluralism:** The Ethics Reasoner does not privilege any single ethical tradition. It is trained to present utilitarian, deontological, virtue-ethical, care-ethical, and contractualist perspectives, identify where they agree and where they conflict, and let the user make the final judgment. This guards against the system becoming a tool for moral indoctrination.
- **Transparent Reasoning:** The system is designed to show its work. Every ethical verdict is accompanied by the chain of reasoning that produced it, making the system's logic auditable and challengeable. If the user disagrees, they can push back, and the dialogue agent will genuinely engage with the objection rather than deferring to authority.
- **Proportional Action:** The Action Agent is constrained to generate only lawful, proportional, and constructive countermeasures. It will not generate malicious code, doxxing tools, or harassment campaigns. Its outputs are limited to legitimate channels: regulatory complaints, consumer awareness, defensive browser tools, and educational content. This constraint is enforced both through fine-tuning and through explicit guardrail prompts.
- **Open-Source Accountability:** By building on open-source models and making the system's code available, the project invites scrutiny. Anyone can inspect the fine-tuning data, the system prompts, and the guardrail logic. This stands in deliberate contrast to the opacity of proprietary AI systems whose ethical commitments cannot be independently verified.

Ethical Implementation: Ensuring Just Use

For this system to be deployed responsibly beyond the classroom, several implementation safeguards would be necessary. First, a clear terms-of-use framework would establish that the system's ethical analyses are advisory, not authoritative—the system assists reasoning but does not replace professional legal or ethical counsel. Second, a robust feedback mechanism would allow users to flag instances where the system's reasoning is biased, incomplete, or harmful, enabling continuous improvement of the fine-tuned models. Third, the Action Agent's outputs would be subject to a human-in-the-loop confirmation step: the system proposes a countermeasure, explains why it is proportional and ethical, and the user must explicitly approve before any action is executed. This prevents the system from unilaterally taking actions that might have unintended consequences.

Personal Ethical Framework

My personal ethical framework for AI development rests on three pillars: transparency, empowerment, and humility. Transparency demands that an AI system's reasoning be legible to its users. Empowerment demands that the system enhance human agency rather than replace it—it should make people better moral reasoners, not dependent on algorithmic verdicts. Humility demands that the system acknowledge the limits of its own reasoning and the irreducible complexity of ethical life. This project embodies all three pillars. It shows its reasoning. It keeps the user in the driver's seat. And it presents multiple philosophical perspectives precisely because no single framework has a monopoly on moral truth. In doing so, it aspires to the standard of decency we have examined throughout this course: an AI that treats every user as a full moral agent deserving of respect, dialogue, and the tools to act on their own convictions.

Appendix: System Architecture Diagram

The diagram below illustrates the high-level architecture of the agentic ethics reasoning system. The user's voice input enters through a speech-to-text module, is routed by the Orchestrator Agent to the appropriate specialized agent (Dialogue, Ethics Reasoner, or Action), and the final output is delivered back to the user via text-to-speech. Ethical guardrails operate as a cross-cutting concern, filtering all outputs before they reach the user. A feedback loop allows the system to refine its analysis through iterative conversation.

[See accompanying architecture diagram image]

Current Status & Next Steps

As of this submission, the following milestones have been achieved or are in progress:

- System architecture designed and documented (complete)
- Technology stack selected and compatibility verified (complete)
- Ethical framework and guardrail design articulated (complete)
- Candidate HuggingFace models identified for evaluation (in progress)
- Local model serving environment setup with Ollama (in progress)
- Voice pipeline prototype with Whisper integration (upcoming)

The final deliverable may evolve from this initial plan—that is a natural and expected part of the development process. However, the core commitments—multi-agent ethical reasoning, voice-first interaction, local privacy, and transparent philosophical pluralism—will remain constant.